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HydraR: Agentic Infrastructure for Reproducible Scientific Research

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Executive Summary

Large Language Models (LLMs) are catalyzing a paradigm shift in scientific computing, fundamentally transforming how research is performed. This transformation is particularly evident in the field of bioinformatics, where LLMs have evolved beyond simple chatbots into autonomous "agents" that can plan and execute complex tasks that previously demanded computational expertise, and powerful multi-agent systems, where a team of AI agents can autonomously collaborate with one another to tackle multi-faceted problems. By democratizing access to these sophisticated multi-agent systems, this automation not only accelerates biological discovery but also empowers the broader scientific community to harness high-performance workflows once restricted to deep domain specialists.

Yet despite offering many great benefits, the current R ecosystem lacks a robust framework for traceable, stateful, and secure agent orchestration. Because of this, multi-agent systems poses a significant risk to the reproducibility of R-based scientific research. Thus, to safeguard the integrity of R-based research, a robust framework must exist to support the development and reproducible use of agentic systems.

To address this critical need, this project will build upon **HydraR**, an existing R package dedicated to the scientific reproducibility of agentic systems. Specifically, this project aims to deliver a robust framework for traceable, stateful and secure agent orchestration. To accomplish this, we will develop a comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions into auditable records for peer review and regulatory compliance. We will also develop a native container layer to ensure agents can iterate and fail safely without risking the integrity of sensitive research environments. Thus, by serving as the central hub for traceable, stateful and secure agentic orchestration, HydraR will empower computational biologists and the broader R community to adopt agentic systems with confidence, and build sophisticated, transparent and reproducible AI-driven workflows that uphold the rigor and integrity of R-based scientific research.

Goals

This project aims to transform HydraR into a foundational infrastructure hub that standardizes agentic workflows for the R community, by developing a production-grade framework that treats

agent workflows as verifiable, recoverable sessions. We aim to empower R users to deploy complex multi-agent systems with the same level of auditable precision expected in traditional computational pipelines.

Methods

We will leverage high-performance technologies, including DuckDB for low-latency state management and native serialization for fault-tolerant checkpointing. Our approach incorporates cryptographic hashing for data provenance and "autopoietic" self-correcting network flows to ensure autonomous research remains within safe, validated boundaries.

Expected Outcomes (Deliverables?)

1. **HydraR R Package**: A comprehensive toolkit for building stateful agentic graphs.
2. **Resilience Layer**: A native checkpointing system to minimize failure costs.
3. **Audit Logs**: Structured, SQL-queryable records of all agent reasoning and tool interactions.
4. **Community Resources**: Documentation and vignettes demonstrating reproducible workflows for the R Consortium.

Budget

Total requested budget: **$14,400 USD**. We request funding for expert software developers to work on this project over a 10-week period (approximately 360 hours).

Executive Summary

HydraR is a high-performance R infrastructure designed to bring scientific rigor and reproducibility to the field of Large Language Model (LLM) orchestration. As autonomous agents become central to bioinformatics research, the R community lacks the tools to ensure these systems are traceable, stateful, and secure. HydraR addresses this gap by implementing a robust "Agentic Infrastructure" layer.

Signatories

Project team

- **Chi Nam (Ignatius) Pang** (ORCID: 0000-0001-9703-5741). **Co-Primary Investigator**. Senior Bioinformatician at Macquarie University / Australian Proteome Analysis Facility (APAF). Expert in R package development, proteomics pipelines, and reproducibility standards.
- **Aidan Tay** (ORCID: 0000-0003-1315-4896). **Co-Primary Investigator (Co-CI)**. Bioinformatician at APAF. Architect of the HydraR R6 DAG engine, Git Worktree isolation logic, and the custom state-hashing persistence Layer.

Contributors

- **Will Klare** (ORCID: 0000-0002-1490-9704). **Associate Investigator**. Bioinformatician at APAF. Expert in reproducible research, statistical methodology, agentic systems, and R package development.
- **APAF Bioinformatics Team**: Provided the testing ground for initial prototypes in high-throughput proteomics and metabolomics research.

- **Antigravity AI**: Agentic assistant used during the rapid prototyping and documentation hardening phase of the project (as disclosed in `agents.md`).

Consulted

- **rOpenSci Editorial Board**: Initial discussions regarding HydraR's role in the AI-R ecosystem.
- **Bioplatforms Australia (BPA)**: Consulted on the impact of agentic orchestration for national-scale omics infrastructure.

The Problem

Large Language Models (LLMs) are catalyzing a paradigm shift in scientific computing, fundamentally transforming how research is performed. This transformation is particularly evident in the field of bioinformatics, where LLMs have evolved beyond simple chatbots into autonomous "agents" that can plan and execute complex tasks that previously demanded computational expertise. This has culminated into powerful multi-agent systems, where a team of AI agents can autonomously collaborate with one another to tackle multi-faceted problems. By democratizing access to these sophisticated multi-agent systems, this automation not only accelerates biological discovery but also empowers the broader scientific community to harness high-performance workflows once restricted to deep domain specialists.

Yet despite offering many great benefits, the current R ecosystem lacks the necessary framework for traceable, stateful, and secure agent orchestration. Existing R packages such as `ellmer` and `mall` provide excellent interfaces to LLM providers to send prompts and parse responses but fail to record the entire decision-making process of an agent, making the "journey" ephemeral and untraceable. Other packages such as `LLMAgentR` and `puppeteeR` enable agent orchestration, but lack the infrastructure for state management, making it impossible for multi-agent systems to meet the rigorous reproducibility standards of modern bioinformatics. Meanwhile, advanced frameworks such as `LangGraph` and `AutoGen` in the Python ecosystem lack the necessary features for auditable agent orchestration, rendering them unsuitable for R-based research via Python-based bridges (i.e., `reticulate`). Moreover, granting autonomous agents unchecked access to research environments can be dangerous as their actions can inadvertently lead to file modifications, deletions, or corruption of the working directory. Thus, to safeguard the integrity of R-based research, a robust framework must exist to support the development and reproducible use of agentic systems.

To address this critical need, this project will build on HydraR, an existing R package dedicated to the scientific reproducibility of agentic systems. Although HydraR provides a solid foundation for memory, state and environment management, it currently lacks several critical capabilties required for production-grade bioinformatic pipelines. Specifically, it does not yet support automated retries for failed tasks, execution caching for skipping duplicated tasks and resource optimization, or workflow validation for autonomously diagnosing and fixing errors. Furthermore, native integration with essential infrastructure tools such as Docker, Git worktrees, and Singularity/Apptainer for High-Performance Computing (HPC) environments, for safely and securely orchestrating agentic workflows is currently absent.

To bridge these gaps, we aim to deliver a robust framework for traceable, stateful and secure agent orchestration. To accomplish this, we will develop a comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions into auditable records for peer review and regulatory compliance. We will also develop an environment management system to ensure

agents can iterate and fail safely without risking the integrity of sensitive research environments. Thus, by serving as the central hub for traceable, stateful and secure agentic orchestration, HydraR will empower computational biologists and the broader R community to adopt agentic systems with confidence, and build sophisticated, transparent and reproducible AI-driven workflows that uphold the rigor and integrity of R-based scientific research.

The Proposal

Overview

The current R ecosystem lacks the necessary framework to ensure integrity of R-based scentific research using multi-agent systems. To address this critical gap, this project will build on HydraR, an existing R package dedicated to the scientific reproducibility of agentic systems. Specifically, this project aims to extend HydraR's functionality by delivering a robust framework for traceable, stateful and secure agent orchestration. In doing so, HydraR will empower computational biologists and the broader R community to adopt agentic systems with confidence, and build sophisticated, transparent and reproducible AI-driven workflows that uphold the rigor and integrity of R-based scientific research.

Details

This project aims to extend HydraR's functionality by delivering a robust framework for traceable, stateful and secure agent orchestration. In particular, this project has two objectives:

Objective 1: To develop a comprehensive memory and state management system for logging ephemeral agent reasoning and agent interactions into auditable records for peer review and regulatory compliance.

Rationale: Agent memory is often ephermeral and untraceable, making it difficult to capture and audit the entire "journey" of an agentic workflow. This poses a significant risk to the reproducibility of R-based scientific research. to ensure that agents are resilient against system and API failures.

Approach: To develop a comprehensive memory system, we will introduce several enhancements to HydraR's memory system:

- **Restart Checkpointing**: We will implement a native serialization engine that maps directly to an `on_failure` lifecycle hook. Every node in the agentic graph will automatically serialize its environment, local variables, and current reasoning state. In the event of a failure (API timeout, resource exhaustion, or logic error), HydraR can reconstitute the exact state of the R session, allowing for manual intervention or automated retries without restarting the entire pipeline.

- **Intelligent Hashing and State Retrieval**: We will implement a content-addressable storage (CAS) model for all agent-generated artifacts. By hashing output files and R objects, we link every piece of data to the specific state and reasoning path that produced it. This enables high-performance state retrieval: the system can "skip" already-computed nodes if the input hashes match, effectively providing a "make"-like capability for agentic workflows.

- **Dynamic Network Repair**: We will implement "autopoietic" self-correcting network flows. If an agent's output fails a validation layer (e.g., malformed code or incorrect data schema), the orchestration engine can dynamically inject a "repair" sub-graph. This sub-graph tasks a

specialist agent with diagnosing and fixing the error, ensuring the main research flow continues safely and autonomously.

- **Network Preparedness and Validation**: The infrastructure will include a "preparedness" layer that governs the growth and pruning of the agent network. Before a new tool or agent is integrated into a workflow, it undergoes a validation pass to ensure its schema and permissions are compatible with the existing security posture. This prevents "state drift" and ensures the network remains optimal and secure.

Objective 2: To develop a native container layer to ensure agents can iterate and fail safely without risking the integrity of sensitive research environments.

Rationale: Security and reproducible environments are paramount for computational biology.

Approach: To address this, we will introduce several enhancmenets to HydraR's container layer to ensure secure and reproducible environments for multi-agent systems.

- **Docker Integration with Git Worktrees**: We will implement a native Docker orchestration layer where each agent execution environment is isolated within a container. To ensure strict traceability of filesystem modifications, HydraR will utilize Git worktrees inside these containers. Every task assigned to an agent will be executed within a unique worktree, allowing HydraR to capture a "git-diff" of all file changes made by the agent. This prevents accidental corruption of the main research branch and provides a cryptographic record of every file modification.

- **Singularity and Apptainer for HPC**: Recognizing that many bioinformatics workflows are hosted on institutional HPC clusters, HydraR will provide first-class support for Singularity and Apptainer. This ensures that agents can be deployed in environments where Docker is restricted. We will tailor this support to standard HPC structures (e.g., SLURM integration and shared filesystem mounts), ensuring that the agentic infrastructure respects the security and resource policies of high-performance computing centers.

- **Persistent State Tracking via Mounted Worktrees**: To bridge the gap between container volatility and workflow statefulness, HydraR will manage Git worktree logic within persistent Docker mounts. By mounting the worktree directory from the host to the container, we ensure that the agent's filesystem state is preserved across container restarts. Combined with our `on_failure` checkpointing, this allows an agent to resume from the exact filesystem state it was in before a crash, while maintaining a versioned history of its progress via Git.

Timeline

Start-up Phase

- **Reporting Framework**: Establish a bi-weekly internal review process to track milestone progress.

Technical Delivery

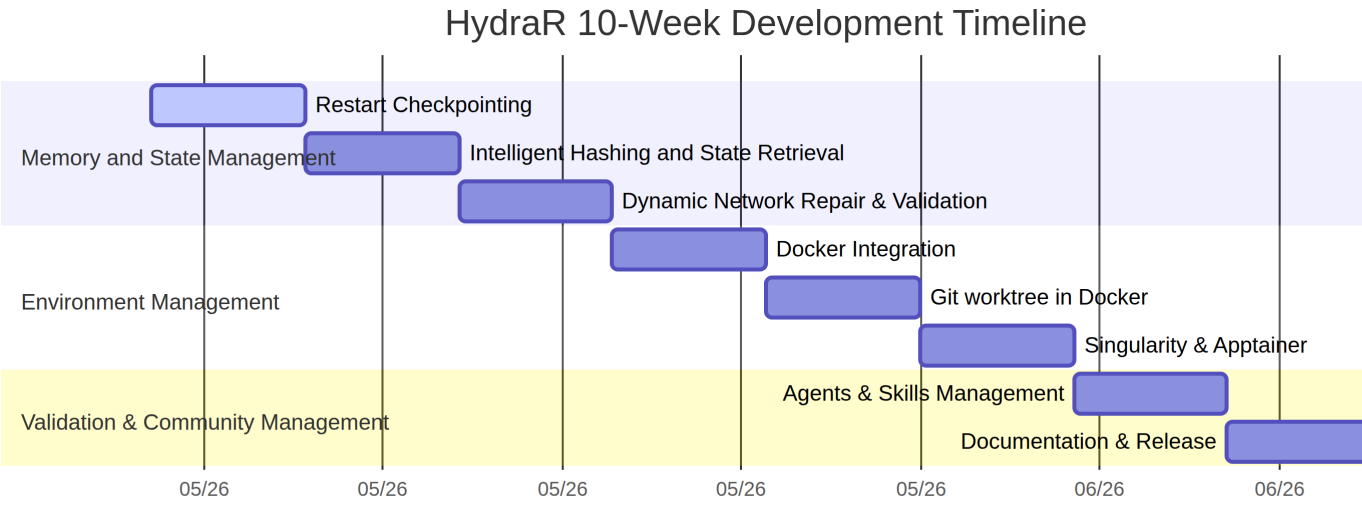
The proposed work will be delivered over a 10-week period and involve approximately 360 hours of expert developer time. The work is structured into three phases:

- **Phase 1: Memory and State Management (Weeks 1-4)**:

  - Development of automated state persistence and restart checkpointing.
  - Development of the serialization engine.
  - Development of dynamic sub-graph injection for self-correcting flows.
  - Development of the validation and pruning layers for network preparedness.

- **Phase 2: Environment Management (Weeks 5-8)**:
  - Development of Docker Integration
  - Development of Git worktree for state tracking in containers
  - Development of compatibility mapping for Singularity and Apptainer

- **Phase 3: Validation & Community Management (Weeks 9-10)**:
  - Finalization of Agents and Skills Management tools into lifecycle modules.
  - Documentation, bug fixing, and release processing.



Other Aspects

- **Hosting**: Code will be hosted on GitHub under the `APAF-bioinformatics` organization.
- **Conference**: We will attend and present HydraR at relevant conferences to engage with the bioinformatics community.
- **Social media**: We will utilize social media to engage with the bioinformatics community.
- **Publicity**: We will share quarterly updates on the R Consortium blog.
- **Community Engagement**: We will share undergo the rOpenSci peer-review process and transition to a community-maintained model.

Budget & Funding Plan

Total requested budget: **$14,400 USD**. We request funding for expert software developers to work on this project over a 10-week period (approximately 360 hours).

Milestone	Description	Estimated Hours	Funding
Memory and State Management	[Details of Milestone 1]	140	$5,760
Environment Management	[Details of Milestone 2]	140	$5,760
Validation & Community Management	[Details of Milestone 3]	80	$2,880
Total		**360**	**$14,400**

| **R Consortium** | **Macquarie University** |
| **Cash (USD)** | **In-kind** |
:— | —: | —: |
Aidan Tay (1.0 FTE) | $14,400 | - |
Chi Nam (Ignatius) Pang (0.2 FTE) | - | $3,431 |
Total | **$14,400** | **$3,431** |

Success

The success of the HydraR infrastructure project will be measured by its adoption, technical stability, and contribution to the R ecosystem's AI readiness.

Success Metrics

- **Package Adoption**: At least 3 existing R/Bioconductor packages demonstrating "Skill Node" integration via the HydraR manifest protocol.
- **Reliability Benchmark**: A published reproducible benchmark dataset showing a >20% improvement in agent task completion rates when using HydraR state management vs. stateless alternatives.
- **Community Engagement**: Successful completion of the rOpenSci peer-review process and transition to a community-maintained model.
- **CRAN/Bioconductor Presence**: Maintain the 0-error/warning/note status across all major R versions.

Definition of Done

The project will be considered successful upon the delivery of:

- A stable CRAN-ready (or Bioconductor-ready) R package named `HydraR`.
- A fully functional "Checkpoint & Resume" engine demonstrated in a multi-agent bioinformatics workflow.
- The capability to perform SQL queries over agent reasoning history via an integrated DuckDB backend.
- Comprehensive documentation, including at least three "reproducible case studies" (vignettes) showcasing traceable agent orchestration.

Measuring success

We will use the following markers to track our progress:

- **Week 2**: Completion of the `duckdb` backend and successful hashing of R objects in a session.
- **Week 4**: Passing unit tests for state serialization and automated recovery from simulated API failures.
- **Week 6**: Successful community beta-test with at least 5 bioinformatics researchers.
- **Week 8**: Final project report and publication of the demonstration blog post on the R Consortium website.

Future work

HydraR is backed by **APAF Bioinformatics** (at Macquarie University), ensuring long-term institutional support and alignment with national-scale bioinformatics infrastructure (NCRIS).

- **Commercial & Institutional Support**: Ongoing maintenance is built into the APAF digital transformation roadmap (2026-2030).
- **Open Source Community**: By providing an interoperable manifest protocol, we encourage a "plugin" ecosystem where community members can contribute drivers and skill nodes without needing deep knowledge of the core DAG engine, ensuring the framework outlives its original funding period.

HydraR is designed to be the foundation for an "Autopoietic Agentic Ecosystem" in R. Future development directions include:

- **Decentralized Orchestration**: Extending the infrastructure to support peer-to-peer agent coordination across distributed R nodes.
- **Advanced Containerization**: Tighter integration with Docker/Singularity for "Secure Execution Enclaves" where agents can test code in isolated environments.
- **Agentic Benchmarking**: A standardized suite for benchmarking LLM performance on bioinformatics-specific R tasks using the HydraR traceability layer.